

Two Ways a Design Fails: When Does a Synthetic LLM Panel See the Danger?

A Backend-Sensitivity Audit of Synthetic-User Interface-Content Risk Measurement

Anonymous Author(s)

Abstract

An AI-advice interface can fail in (at least) two ways: it can make human reliance *heterogeneous and user-sensitive* (an *over-dispersion* failure), or make users *over-rely on a confidently wrong AI* under a coercive framing (a *dark-pattern* failure). Detecting either normally needs an expensive human study, so a tempting shortcut is to screen the interface first with a multi-agent LLM panel playing a synthetic cohort of users. Most debate about such panels asks whether they *match* humans; we ask a question that comes first and is rarely tested: is the panel's risk *measurement* even stable across the choices a researcher makes freely when building it—the backend model, the persona prompt, and the generation sampling? Auditing this across six backend LLMs and two review-sentiment datasets, we separate *what is stable from what is not*. **Unstable:** the aggregate risk estimate for the *same* interface swings widely with the backend (wrong-advice adoption 0.15–0.66), enough to *flip* the interface's risk classification for a majority of backend pairs (pairwise flip rate up to 0.60); a matched contrast confirms a real coercive-framing effect (pooled item-clustered odds ratio 1.90 on one dataset, 1.17 on the other; a non-coercive placebo at or below baseline) of a magnitude we cannot distinguish across backends—so the backend moves the adoption *level*, not the framing effect. **Stable:** the panel faithfully propagates a *prompted* disposition—an explicitly trusting-novice persona is the highest-adopting persona in all 12 backend×dataset cells (crossed random-effects odds ratio ≈ 26)—a reliability property of the simulation, not a discovered vulnerability; a variance decomposition confirms per-trial adoption is driven *more* by the researcher-chosen items and personas than by the backend, which nonetheless still moves the aggregate estimate a practitioner reads. **Most consequential:** the strongest backend is the one that least reproduces the highest-severity failure—the at-risk trusting-novice persona—so a single frontier backend (the model a quality-optimal router would pick) surfaces almost none of the risk that a pair of weaker backends recovers. **Usable, with a caveat:** used to *compare* designs rather than certify one, the panel is more dependable—a protective framing (cognitive forcing or a verification display) significantly lowers synthetic over-reliance, most for the at-risk persona, and every backend reproduces the *direction* of that improvement, though not its magnitude. We contribute (i) a systematic backend-sensitivity audit showing that synthetic-user interface-risk measurement is not backend-invariant; (ii) a *capability–vulnerability mismatch* and a vulnerability-coverage principle for panel selection—the inverse of capability routing; (iii) a reusable audit and reporting protocol (variance decomposition, behavior taxonomy, cross-backend decision-flip, disagreement-triggered abstention)—report distributions over models/prompts/generations, not a point estimate; and

(iv) the two-axis over-dispersion/coercion construct on a preregistered, audit-gated pipeline. We make *no* claim that the panel predicts individual humans or identifies a real vulnerable population; human predictive validity is reserved for a separate, later study. The result is an honest map of *when a synthetic panel's danger reading can and cannot be trusted*—before any human is recruited.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; • **Computing methodologies** → Artificial intelligence.

Keywords

human–AI decision making, over-reliance, dark patterns, LLM simulation, synthetic users, pre-deployment evaluation, silicon sampling

1 Introduction

Consider a product team about to ship a review-triage tool. The interface shows an AI's verdict and a confidence score and—to feel trustworthy—a highlighted “explanation.” Averaged over a pilot, the numbers look fine: accuracy is up, users seem to like it. What the average hides is *who* the interface endangers and *when*. A cautious domain expert barely moves; a *trusting novice* takes the AI's word almost every time—and if a coercive variant presents a confidently wrong verdict, that same profile may be steered toward the wrong decision while the aggregate barely moves. The design has not failed on average. It has failed for a person, in a way the mean cannot see.

This is the gap between *mean accuracy* and *safety*. Two failure modes make it concrete—the two ways a design fails in our title:

- **Axis 1—over-dispersion.** Reliance becomes *heterogeneous across users* in a way the interface induces. The team ships something safe for the median user and unsafe for the tails. Variance, usually a nuisance to be averaged away [3, 10], is here the *safety quantity of interest*.
- **Axis 2—wrong-advice adoption under coercion.** Under a coercive framing, users *systematically adopt a wrong AI*. This is dangerous even at zero dispersion: everyone can fail together [6, 15, 31].

The standard way to detect either failure is a human-subjects study—powered, IRB-approved, and slow. Teams iterate on interfaces far faster than they can run such studies, so a growing practice is to screen an interface first with a **synthetic panel**: persona-conditioned LLM agents that role-play a cohort of users, from which the two axes are estimated. Synthetic users are having a moment in HCI [1, 2, 20], and so is the backlash: a large literature warns that LLM-simulated people are *unreliable proxies*—they collapse human diversity, echo the base model's priors, and mis-estimate distributions [4, 7–9, 12, 23].

This is no longer hypothetical. A commercial market of synthetic-user research platforms now advertises AI “participants” that evaluate designs *in minutes* and compress—or stand in for—weeks of human study [28, 30]. Such a practice quietly rests on two assumptions that, to our knowledge, are never tested for interface-risk evaluation: (i) that the *strongest available* backend is the best simulator—the natural engineering default, and exactly what a quality-optimal model router would pick [19]; and (ii) that a *single* backend’s point estimate (“this design’s wrong-advice adoption is 0.6”) is a property of the design rather than of the model that produced it. This paper puts both assumptions to an empirical test, and finds both wanting.

Our stance is neither boosterism nor rejection. Simulating users with LLMs is a reasonable and advanced way to cut the cost of early evaluation, and we treat it as such. But a practitioner should not trust its *single number*. Crucially, our caution is not the usual “the agents are not human enough”—that would require a human ground truth we (and the field) largely lack. It is a *measurement-reliability* caution that needs no human criterion: swap the backend model, and the *same* interface can cross from “safe” to “dangerous.” Rather than abandon the method, we show how to use it honestly—power a panel with *several capability-diverse* backends chosen to cover the failure modes a single strong model misses, report the risk as a *distribution* rather than a point estimate, and *abstain*—routing the design to a human study—exactly when the backends disagree. Cheaper than an all-human study, but not credulous.

That debate is almost entirely about whether synthetic users *match* real ones. We argue a more basic question comes first, and is rarely tested. When a panel reports that an interface is risky, that reading is produced not only by the interface but by a stack of choices the researcher makes *freely*: which backend LLM powers the agents, how the personas are worded, and which random generation happens to be sampled. If those choices move the reading as much as the interface does, then a “design risk” number may be as much a property of the measurement tool as of the design. Before asking whether a synthetic panel predicts humans, we should ask whether it even *measures the same thing* when a different researcher, or the same researcher on a different day, rebuilds it. This is a question of measurement reliability and invariance [8, 23]—logically prior to external validity—and it is the question this paper answers.

Scope. Throughout, “interface” refers to the interface’s *informational content and framing* as an agent receives it—the AI prediction, its confidence, the highlighted explanation, and the coercive or protective wording—encoded as text (Section 4). We deliberately study this content/framing layer, not the pixel-level visual layer (layout, salience, typography); a multimodal rendering is future work (Section 7). Our claims concern whether a synthetic panel’s reading of *that* content is backend-stable, not the visual presentation of a shipped UI.

Concretely, we audit a two-axis synthetic panel across **six backend LLMs** and **two review-sentiment datasets**, and ask:

- **RQ1 (backend invariance).** Do the panel’s interface-risk estimates—absolute wrong-advice adoption, the coercive-framing effect, and persona heterogeneity—agree across backends, and can a backend change *flip* an interface from “high risk” to “low risk”?

- **RQ2 (variance decomposition).** How much of a synthetic risk estimate is attributable to the backend model versus the researcher-chosen items and personas versus generation randomness?
- **RQ3 (construct vs. compliance).** Which panel signals reflect a stable sensitivity to the intended manipulation, and which primarily reflect explicit persona instructions?
- **RQ4 (backend-aware reporting).** Can conservative multi-model aggregation and disagreement-triggered abstention reduce backend-induced instability in the measurement?

Our answer separates *what is stable from what is not*. **Not stable (RQ1):** the aggregate risk estimate for the same interface swings from 0.15 to 0.66 across backends (Figure 1), enough to flip the interface’s risk classification for a majority of backend pairs across a wide range of thresholds (Section 5.4); a matched contrast that holds the wrong advice constant and varies only the framing confirms a real coercive-framing effect (pooled item-clustered odds ratio 1.90 on one dataset, 1.17 on the other; a non-coercive placebo at or below baseline; Section 5.3), of a magnitude we cannot distinguish across backends—so it is the adoption *level*, not the framing slope, that the backend moves. **Stable, but for an unglamorous reason (RQ3):** the panel faithfully propagates a *prompted* disposition—an explicitly trusting-novice persona is the highest-adopting persona in all 12 backend×dataset cells (crossed-mixed-model odds ratio ≈ 26 ; Figure 4)—which is a reliability property of the simulation, not a discovered vulnerability, since the persona is defined to be trusting; and its cross-vendor *ordering* agreement is largely carried by this one persona (Section 5.5). **Sources (RQ2):** per-trial, the researcher-chosen items and personas explain more variance than the backend, yet the backend still moves the aggregate estimate a practitioner reads (Section 5.5). We use these to motivate a backend-aware reporting protocol (RQ4). **Most consequential:** backend capability and reproduced vulnerability are *mismatched*—the highest-capability backend least reproduces the highest-severity failure, so the single frontier model a quality-optimal router would choose surfaces almost none of the risk that a pair of weaker backends recovers (Section 5.7); selecting a panel for *vulnerability coverage* is therefore the inverse of capability routing. **Usable, with a caveat:** when the panel is used to *compare* designs rather than certify one, it is more dependable—adding a protective framing (cognitive forcing or a verification display) to a wrong-AI interface significantly lowers synthetic over-reliance, most for the at-risk persona, and every backend reproduces the *direction* of that improvement (Section 5.8)—though not its magnitude (on one backend the safeguard is inert), and, against our preregistration, the benefit is not the frontier that is resistant.

We keep the evocative “two ways a design fails” framing because the two axes are the risks we care about; but our *claimed* contribution is the audit, not a validated screen. We contribute:

- (1) **A systematic backend-sensitivity audit** of synthetic-user interface-risk measurement (six backends × two datasets), organized around a *risk-classification flip rate*—the fraction of backend pairs that disagree on whether the *same* interface crosses a decision threshold (up to 0.60). The aggregate risk estimate is strongly *not* backend-invariant while a prompted persona ordering propagates stably. Prior

prompt/model-sensitivity work shows outputs *shift in magnitude*; we show the shift *crosses a decision-relevant threshold*, flipping a pass/fail verdict—measurement (non-)invariance made decision-theoretic, which is what makes it consequential for a deployment gate.

- (2) **The capability–vulnerability mismatch and a vulnerability-coverage principle for panel selection.** The strongest backend is the one that least reproduces the highest-severity failure (the at-risk trusting-novice persona), so a single frontier backend—the model a quality-optimal router [19, 32] would pick—surfaces almost none of the risk that a pair of weaker backends recovers. Choosing an evaluation panel to maximize *vulnerability coverage* is thus the *inverse* of capability routing: a counterintuitive, actionable design principle for who should power a synthetic-user study.
- (3) **A reusable audit and reporting protocol:** a variance decomposition over backend, persona, item, and generation; a behavior taxonomy; a cross-backend decision-flip analysis; and disagreement-triggered abstention—report synthetic-panel results as *distributions over models, prompts, and generations*, not a single point estimate, and abstain when backends disagree.
- (4) **The two-axis over-dispersion/coercion construct**, conflict-conditioned and preregistered on an audit-gated pipeline whose process integrity (three self-caught silent-corruption bugs; Section 4) is part of the method.

We make *no* claim that the panel predicts individual humans, identifies a real vulnerable population, or constitutes a validated deployment screen; our panel↔human correspondence is null in every (underpowered) configuration we can currently test. Human predictive validity is reserved for a separate, later study (Section 8); this paper establishes the internal robustness and construct sensitivity that must precede it.

2 Related Work

Simulated users, and the reliability critique. LLMs as stand-ins for human participants have been proposed for surveys, user studies, and formative evaluation [1, 2, 20]. A parallel literature documents their failure as proxies: they compress opinion diversity and over-represent majority views [23], mis-calibrate distributions [7, 9], exhibit non-human response biases [29], fail as survey-sample replacements [4], and caricature rather than simulate subpopulations [8, 12]. We sit inside this debate but change the target: we do not ask the panel to *predict* a human distribution, we ask it to *triage* an interface, and we isolate a specific, measurable pathology—*model-dependence of the triage signal*—that the prediction framing obscures. Two recent efforts sharpen this contrast. Grounding agents in real interviews, Park et al. [21] report LLM agents that reproduce individuals’ survey responses with ≈85% accuracy—an *accuracy/correspondence* result; we ask the logically prior *reliability* question (is the risk reading even reproducible across backends?) that such a validation effort presupposes. Hu and Collier [13] decompose how much annotation variance a *persona* explains and find larger instruction-tuned models simulate personas *better* where persona matters; our decomposition instead isolates the *backend* as the free variable and, in the interface-risk setting, finds the *opposite*

for the highest-severity persona (Section 5.7)—a domain-specific tension we name the capability–vulnerability inversion rather than a contradiction.

Reliance and over-reliance on AI advice. A large body of work studies when people appropriately rely on AI given confidence and explanations [3, 6, 14, 24, 33], often finding explanations *increase* over-reliance rather than calibrate it [3, 31]; recent work benchmarks reliance *interventions* (disclaimers, uncertainty highlighting) with human participants on tasks including LSAT and finds they curb over-reliance but rarely improve *appropriate* reliance [5]. This work treats between-user variance as noise around an average, and studies humans directly. We invert the first: reliance *heterogeneity* is our axis-1 safety DV, and coercive wrong-AI adoption is axis-2; and we complement the second: our protective-intervention arm (Section 5.8) asks not *whether* an intervention helps humans but *whether a synthetic panel can even detect that it helps*, and how that verdict depends on the backend.

Dark patterns and coercive interfaces. Deceptive and coercive designs are well catalogued [15, 16]; our axis-2 gives a computational signal for one coercive mode (a confidently wrong AI) and, via personas, a read on which prompted profiles the simulation marks as susceptible.

Offline evaluation of human–AI teams. Closest is offline estimation of team performance for decision support [22]; that work studies bias and complementarity, not interface-level safety triage. The gap we fill is a *pre-study screen for interface reliance safety*, and—critically—a characterization of when such a screen’s signal is trustworthy.

Measurement invariance, reliability, and validity. Psychometrics distinguishes *reliability* (does an instrument give the same reading on repetition) from *validity* (does it measure the intended construct), and *measurement invariance* asks whether an instrument means the same thing across groups or contexts [23, 29]. We import this lens: the *backend LLM is part of the measurement instrument*, so comparing interface risk across backends is a measurement-invariance question, and stability across generations is a reliability question. Stability does not imply validity, but severe backend non-invariance undermines any validity claim built on a single backend [9].

Researcher degrees of freedom in LLM evaluation. Model choice, prompt wording, decoding settings, output parsing, persona phrasing, and sample count are analytic choices that can each move a reported result—a form of hidden flexibility analogous to the “garden of forking paths” and undisclosed analytic flexibility in empirical research [26]. In LLM evaluation specifically, small prompt-format changes swing results [25] and single-prompt scores misrepresent ability [18]; a generalizability-theory decomposition of LLM output variance into model, prompt, and sampling components has been applied to creative tasks [11]. We make these degrees of freedom the object of study, and adopt specification-curve-style multiverse reporting [27] rather than a single spec. Our decomposition targets a different quantity than this prior work—not the magnitude or quality of an output, but whether a *binary interface-risk verdict* survives the researcher’s free choices.

Algorithmic auditing and robustness reporting. Our stance—stress-test a widely-adopted measurement practice, quantify its sensitivity, and recommend distributional reporting with abstention—follows the tradition of sensitivity analysis, model/evaluation cards [17], and multi-model benchmarking with reliability-aware judging [34]. We contribute a concrete audit protocol for synthetic-user interface evaluation. Our novelty relative to this body of work is the *object* and the *consequence*: prior prompt/model-sensitivity results are about accuracy or benchmark scores, whereas we show that the same free choices move an interface-level *safety decision* in the synthetic-user HCI evaluation setting—flipping whether a design is flagged for a human study—which is the point at which such measurement instability actually matters for design practice.

3 The Two-Axis Construct: Definitions

We characterize an interface on two axes, both measured on the panel’s decisions.

Axis 1—between-user reliance over-dispersion. For each item, personas produce reliance decisions; we estimate the *excess variance* beyond the binomial $p(1 - p)$ a single homogeneous user would produce (a beta-binomial over-dispersion estimator, parameter p). We compute it *conflict-conditioned*: only on trials where the persona’s independent System-1 judgment disagrees with the shown AI, isolating genuine reliance from trivial agreement. As a stable, bounded summary that avoids the beta-binomial estimator’s boundary behaviour at extreme rates (Section 4), we report *mean panel disagreement*: for a condition with personas $j = 1, \dots, P$ ($P = 6$), let r_j be persona j ’s conflict-conditioned reliance rate; the condition’s disagreement is their between-persona (sample) variance

$$D = \frac{1}{P-1} \sum_{j=1}^P (r_j - \bar{r})^2, \quad \bar{r} = \frac{1}{P} \sum_{j=1}^P r_j, \quad (1)$$

and we report the mean of D over the five UI conditions. D tracks p monotonically and is what we plot; we report p alongside it. High axis-1 means the panel estimates reliance as persona-sensitive—a candidate signal of user-sensitive risk, not yet human-validated.

Axis 2—wrong-advice adoption under coercion. Under a coercive “dark” framing, the interface shows a *guaranteed-wrong* AI verdict (the negation of ground truth) at high confidence with authority and accountability cues. Axis-2 is the rate at which personas adopt it. To separate the *framing* from ordinary wrong-advice following, we contrast it against neutral and placebo framings on matched items (Section 5.3). It is dangerous even at zero dispersion—a uniform failure.

From axes to a decision, and why we audit it. A screen must eventually turn the two axes into an action—e.g., “flag for a human study if either axis exceeds a threshold τ ; otherwise treat as a release candidate; abstain when out-of-distribution.” We deliberately do *not* propose or validate such a rule here; instead we take the simplest version of it (a threshold on axis-2 adoption) as the *object of the audit*, and ask whether the decision it produces is stable across backends (Section 5.4). Any thresholds remain unfrozen: their calibration and validation depend on the human study (Section 8).

4 Method: The Synthetic Panel

Panel = personas $\times$ backends. Each agent is a persona (a vector of domain skill, AI literacy, risk sensitivity, caution) crossed with a base LLM (the “backend”). Personas span novice/expert $\times$ trusting/skeptical plus two moderates (six in total). Crucially, a persona’s prompt is generated identically across backends and its decoding temperature is fixed *per persona*, not per backend, so temperature and wording are held constant when we vary the backend—backend effects are not confounded with per-model prompt or temperature differences.

Pre- and post-advice decisions. Each agent first produces a judgment *without* AI (a frozen “System-1”/pre-advice anchor), then a decision *with* the AI advice and framing (“System-2”/post-advice). We use the System-1/System-2 labels descriptively, not as a validated model of dual-process cognition. Reliance is defined against the pre-advice anchor—what makes conflict-conditioning possible and prevents the “everyone-agrees-so-everyone-complies” collapse of a naive panel.

Textual prompt encoding of the interface. Agents receive a *textual serialization* of each interface condition—not screenshots and not a vision model. The AI prediction, its confidence, and (for the explanation conditions) the highlighted tokens are rendered as text in the prompt, following the Bansal et al. [3] condition definitions (confidence-only; single/double highlighted-token explanations; adaptive and expert-adaptive). The dark condition negates ground truth and adds high confidence plus authority/accountability language; a placebo condition presents the real AI non-coercively. This means we study *prompt-level* interface framing; pixel-level layout, visual salience, and typography are not modelled, and a multimodal rendering is future work. The exact per-model prompt templates are in the supplement.

Stimuli and datasets. Items are drawn from the Bansal sentiment benchmark, held out from any threshold fitting, and selected to concentrate on loci of human reliance heterogeneity (disclosed; human outcomes are used only for *item selection*, never leaked into agent prompts—item selection $\neq$ label leakage). We run two *review-sentiment datasets*: **beer reviews** and **Amazon book reviews (amzbook)**. Both are English binary sentiment; we therefore treat this as replication across two datasets within one task family, not broad cross-domain generalization (Section 7).

Backends. A within-provider series (gpt-4o-mini, gpt-4.1, gpt-4o, gpt-5.5, identical config) plus independent-vendor models (claude-sonnet-4.5, gemini-2.5-pro): six backends total. We order the same-provider series by release for presentation only and make no claim that this is a validated capability scale; “frontier” refers to the most recent, most capable general model at the time of writing (gpt-5.5).

Rigor spine and process integrity. Before running each arm we froze—with a UTC-timestamped, committed record—the backend set, persona set, item-selection seeds, conditions, primary DVs, and the predicted direction of each effect; deviations and post-hoc analyses (e.g. the exploratory threshold sweep and the descriptor ablation) are logged and labelled as such. Correspondence analyses were coded blind; every merge was gated on an independent audit and a from-raw numeric re-derivation. This spine caught three

silent-corruption bugs that would each have produced a plausible-but-false result: an axis-2 *sign inversion* (scoring adoption against ground_truth instead of the displayed $1 - \text{ground_truth}$), a beta-binomial *boundary artifact*, and a decision parser that silently *fabricated* a (decision=0, conf=0.5) sentinel on parse failure. It also caught a statistical error in our own draft—a GEE interaction test that appeared to show backend-dependent framing but was an artifact of few item clusters (Section 5.3). We report these openly: a pipeline that catches its own errors is what makes the numbers trustworthy.

5 Experiments and Results

We report per-backend (no pooling across incomparable backends), with effect sizes, confidence intervals, and Benjamini–Hochberg (BH) correction. Unless noted, the BH family is the 12 per-cell (backend×dataset) binomial tests of dark-condition adoption against 0.5. There are $n = 120$ dark-condition trials per backend×dataset cell (6 personas × 20 items).

5.1 Real human over-dispersion exists (anchor)

We first verify that axis-1 corresponds to a measurable human quantity in the source benchmark. On the Bansal et al. [3] human data, we estimate between-user reliance over-dispersion with the same beta-binomial estimator we apply to the panel; conflict-conditioned over-dispersion is positive and excludes zero ($p \approx 0.067$), no-AI reliance behaves as a stable trait (≈ 0.74), and AI-assisted reliance is a user×task interaction (0.32–0.41). We report this anchor briefly (full participant counts, per-condition estimates, and intervals are in the supplement); it establishes that the human construct is measurable but does *not* generalize to a second human dataset (Lu & Yin), which we disclose as a boundary. The human anchor is not used to validate the synthetic panel—that link is null in every configuration we test (Section 5.9).

5.2 Wrong-advice adoption is strongly backend-dependent

Backend choice changes both the *level* and the *interpretation* of wrong-advice adoption (Figure 1). Across the ladder (beer / amzbook): gpt-4o-mini 0.50 / 0.48, gpt-4.1 0.60 / **0.66**, gpt-4o 0.43 / 0.48, gpt-5.5 **0.30** / **0.15**. The adoption peak at gpt-4.1 is real (between-model Fisher vs. gpt-4o $p = 0.014/0.006$) but **model-idiosyncratic**: the smallest ladder model is at chance, and after BH correction gpt-4.1’s elevated adoption survives on amzbook ($p = 0.003$) but is only nominal on beer (uncorrected $p = 0.035$; BH $p = 0.11$). Independent-vendor frontier models are at chance (claude 0.49, gemini 0.53 on beer; 0/3 vendors above chance). Rather than interpret adoption against a 0.5 “chance” baseline—which is not a natural no-effect point, since personas are not random guessers—we ground the effect in a matched framing contrast below.

Axis-1 tells a parallel story (Figure 2). We call this quantity *synthetic persona disagreement*—the between-persona variance of the six personas’ conflict-conditioned reliance rates (Section 3)—and are explicit that it measures how differently our *prompted* personas respond, not a validated human over-dispersion. Mean persona disagreement across the model set is 0.11 / 0.16 / 0.10 / **0.012** (beer)

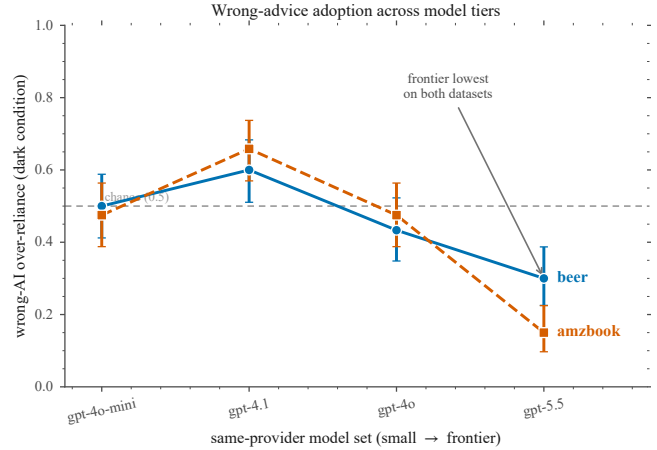


Figure 1: Axis-2 wrong-advice adoption on the guaranteed-wrong interface across the same-provider model set (gpt-4o-mini → gpt-4.1 → gpt-4o → gpt-5.5), both datasets. Error bars are Wilson 95% binomial confidence intervals (descriptive; conflict-conditioned inference uses the item-clustered models below), and lines connect discrete model conditions only as a visual guide; dashed line = 0.5. Adoption is non-monotonic across backends; gpt-5.5 has the lowest adoption on both datasets. The matched framing contrast (Section 5.3) provides a non-arbitrary baseline for this low level.

and 0.12 / 0.19 / 0.11 / **0.015** (amzbook): gpt-5.5 is the most homogeneous on both datasets, and the beta-binomial p (which mean disagreement tracks) is correspondingly near zero. Disagreement is computed over the full 5-condition design (5 conditions × 6 personas × 20 items = 600 trials/model), distinct from the 120 dark-only axis-2 trials; the low value at gpt-5.5 is not a no-conflict artifact, since genuine conflict trials remain present (e.g. 215/600 on amzbook), and it also holds on the common conflict subset shared across models. We caution against reading it as a safety construct collapsing: low disagreement at a strong model can equally reflect the personas converging on the *correct* answer (uniform task-solving) rather than an interface failing to differentiate users—these are indistinguishable without a human criterion.

Reading. gpt-5.5’s cross-dataset low adoption and axis-1 collapse are the most consistent patterns; adoption magnitude is otherwise model-specific. This motivates the decision-instability audit next: if the aggregate reading is this model-specific, does the *decision* a screen would make move with it?

5.3 Is it coercion, or just wrong-advice following? A matched framing contrast

Adoption of a guaranteed-wrong AI could reflect ordinary wrong-advice following rather than the *coercive framing* (high confidence, expert-authority and accountability cues). To separate them we use a within-item contrast: on the subset of items where the AI advice is *naturally wrong* under the neutral condition (8 items/cell), the same wrong label is shown under neutral, non-coercive placebo,

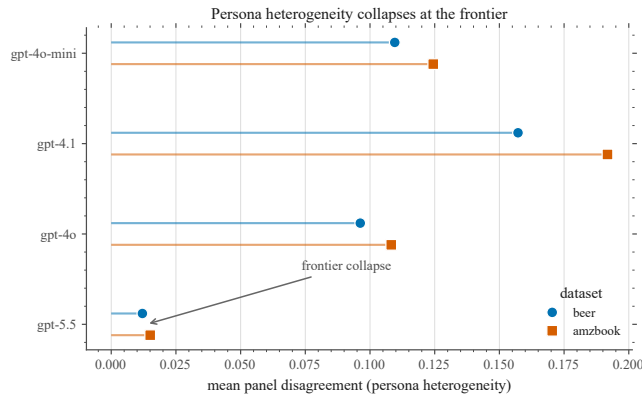


Figure 2: Axis-1 mean panel disagreement across the same-provider model set, both datasets (error bars: SD across the 5 UI conditions). gpt-5.5 shows the lowest cross-persona disagreement on both datasets.

and coercive-dark framings—so any difference in adoption is attributable to the framing, not to the advice being wrong. We fit a logistic model $\text{adopt} \sim \text{dark} + \text{placebo} + \text{model}$ (clustered by item) for the pooled framing effect, and test backend-dependence with a likelihood-ratio test of the $\text{dark} \times \text{model}$ interaction.

A coercive-framing effect is present: pooled across backends, the dark framing raises wrong-advice adoption over the matched neutral baseline with odds ratio **1.90** on beer (95% CI 1.42–2.53) and **1.17** on amzbook, while the non-coercive placebo stays at or below neutral (beer OR 0.94, n.s.; amzbook OR 0.88, $p = 0.005$, a small protective effect)—so the effect is the coercive framing, not mere AI-presence. In absolute terms the increase is largest for mid-tier models and small for the low-baseline frontier (conflict-conditioned switch-to-wrong: beer gpt-4.1 0.19 $\rightarrow$ 0.51, gpt-4o 0.12 $\rightarrow$ 0.29, gpt-5.5 0.04 $\rightarrow$ 0.10). However, a formal test does *not* support a backend-dependent framing effect: the $\text{dark} \times \text{model}$ interaction is non-significant on both datasets (likelihood-ratio $\chi^2(5) = 1.70$, $p = 0.89$ beer; $\chi^2(5) = 3.18$, $p = 0.67$ amzbook). We therefore claim only that (i) a real *bundled* coercive-framing effect exists (confidence, authority, and accountability cues co-vary and are not separated), of a magnitude we cannot distinguish across backends; and (ii) what is strongly backend-dependent is the absolute adoption *level* (Section 5.4), not the framing slope. This also replaces the arbitrary 0.5 baseline: the frontier’s low adoption is a low *level*, measured against a matched neutral condition rather than against chance.

5.4 Cross-backend decision instability: the same interface, flipped

Our first audit question (RQ1) treats the dark interface as one *design* and asks whether backends agree on its risk (Figure 3). They do not. Aggregate wrong-advice adoption for the identical interface is 0.30–0.60 across the six beer backends and 0.15–0.66 across the amzbook backends. Turning the estimate into the binary decision a screen would make—“flag for a human study if adoption $\geq \tau$ ”—the flag is unstable not just at one threshold but across the entire

plausible operating range: as τ sweeps 0.35–0.65 (beer) and 0.30–0.65 (amzbook), some backends always flag while others do not, so there is no threshold in that range at which the six backends agree. At $\tau = 0.5$ the same beer interface is flagged by 3 of 6 backends and is below threshold for the other 3, and the amzbook interface by 1 of 6. The **pairwise backend flip rate**—the fraction of backend pairs that disagree on the flag—peaks at **0.60** on beer and 0.33 on amzbook. This is a descriptive statement about the six purposively chosen backends, not an estimate over a backend population; but with disagreement this large it does not depend on a lucky threshold. Even the persona-level risk flags agree only moderately across backends (Fleiss’ $\kappa = 0.64$ beer, 0.24 amzbook). In measurement terms, the instrument is not invariant: swapping the backend can move an interface from “high risk” to “low risk” without touching the design. A behavior audit rules out one benign explanation—that low-adopting backends simply *refuse* the deceptive task: explicit challenge/refusal language is rare (0% of dark responses for the OpenAI backends, 2–4% for claude and gemini), and low adoption is achieved by evidence-based re-verification, not by declining to answer.

5.5 Where the variance comes from

A variance decomposition explains why backend instability can coexist with even larger item and persona effects at the trial level. A crossed random-effects model of per-trial adoption ($\text{adopt} \sim 1 + (1|\text{backend}) + (1|\text{persona}) + (1|\text{item})$) apportions variance *more* to the researcher-chosen items and personas than to the backend (approximate variance shares: item ≈ 0.28 , persona ≈ 0.23 , backend ≈ 0.07 , residual ≈ 0.42 ; between-group SDs 0.16/0.25/0.13 corroborate the ordering). This is not a contradiction of Section 5.4 but a sharpening of it: at the level of a single trial the backend is a modest source of noise, yet at the level of the *aggregate estimate a practitioner reads* the backend still shifts the number enough to flip the decision. The practical implication is the same for every source—item selection, persona wording, and backend are all analyst choices that move the reading, so a synthetic-panel result should be reported as a *distribution* over these choices, not a point estimate. Generation randomness—the one source a single draw per prompt cannot expose—turns out to be the smallest of all. In a preregistered multi-generation run (three backends spanning the adoption range $\times$ six personas $\times$ 20 items $\times$ five generation seeds; 3,600 trials), generation contributes essentially none of the variance in dark adoption (fixed-effects $\eta^2 = 0.001$, versus 0.22 for items, 0.15 for personas, and 0.07 for the backend), and each backend’s aggregate over-reliance estimate moves by at most SD = 0.03 across the five seeds (flip-metric range ≤ 0.09)—small against the 0.31 spread *between* backends that flips the decision. Generation noise is real at the level of an individual persona–item cell (mean cell SD 0.17; about a third of cells change across seeds), but it averages out—two-thirds of cells are reproduced identically—so the aggregate reading a practitioner acts on is stable across generations, and the backend and persona effects above are not artifacts of a lucky draw. (We report a fixed-effects η^2 share because a random-effects variance component for the backend is degenerate on three groups; the direct stability statistics are the primary, model-free evidence.)

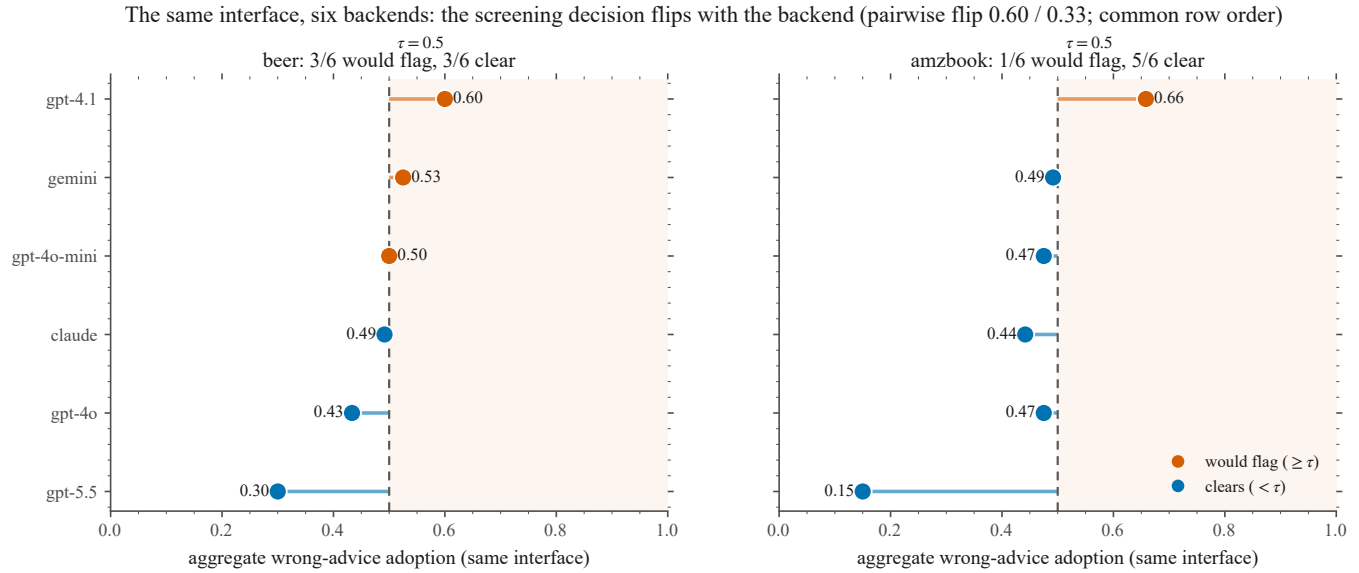


Figure 3: The *same* guaranteed-wrong interface, scored by six backends, both datasets. Bars are aggregate wrong-advice adoption; the dashed line is a fixed decision threshold ($\tau = 0.5$). Red backends are at or above threshold (would *flag* the interface), green backends are below it—so the screening decision flips with the backend (pairwise flip rate 0.60 beer / 0.33 amzbook).

5.6 A backend-stable manipulation check: the panel propagates a prompted trust disposition

The aggregate adoption *rate* is model-idiosyncratic (range 0.30 across beer backends, 0.51 across amzbook backends). The *ordering* of which persona adopts most is stable (Figure 4). An **explicitly trusting-novice** persona is the highest-adopting persona in **all 12** backend×dataset cells. Because the 20 items are reused across personas and the 12 cells share prompts, items, and providers, we do not treat trials as independent: a crossed random-effects logistic model with item, backend, and dataset random effects gives an odds ratio of 26 (95% CI 17–41) for that persona relative to the others (an item-clustered GEE agrees, OR 14.9). The effect is robust to the pseudo-replication structure.

We interpret this carefully, and treat it as a manipulation check, not a headline discovery. The persona is *prompt-defined* to be trusting and novice, so “it adopts the AI most” is substantially a consequence of the instruction, not a discovery. What the result shows is a reliability property: the panel *faithfully and stably* translates a prompted disposition into behaviour. But the cross-vendor *ordering* agreement is largely carried by this one persona: the mean pairwise Spearman across independent vendors is 0.87 on beer with p5 included, but drops to 0.77 on beer and *collapses to 0.09* on amzbook once p5 is excluded (Figure 5)—so beneath the shared top rank the vendors do not agree on the ordering. It is *not* evidence that the panel discovers a latent vulnerable human population; a chance baseline of 1/6 per persona would be inappropriate here, since the personas are not exchangeable (this one was constructed to be the most trusting). A within-data check reinforces this: holding the “novice” background fixed and varying only the trust disposition, the novice-*skeptical* persona (p1) adopts far less than the

novice-*trusting* persona (p5) on every backend—so the effect tracks the explicitly instructed AI-deference policy, not “novice-ness” or a discovered trait. Whether the ordering survives *removing* the explicit deference instruction entirely is a preregistered synthetic *descriptor ablation*: we re-run the dark condition on two backends with background-only persona prompts (experience level and AI-usage frequency stated, but no explicit trust/deference policy). The result is mixed and *backend-dependent*, and we report it as such. On the mid-tier backend (gpt-4.1) the trusting-novice’s adoption is essentially retained without the policy (1.00 with policy vs. 0.90 background-only) and the persona ordering is preserved in point estimate (Spearman 0.70, n.s. at $n = 6$)—but p5 is no longer the *unique* top adopter, because the skeptical personas also become more compliant once their explicit verification instruction is removed (the novice-skeptical persona p1 rises from 0.60 to 0.90). On the frontier backend (gpt-5.5) the signal is more clearly policy-carried: p5 falls from 0.60 to 0.40 and from rank 1 to rank 2, with ordering agreeing only Spearman 0.34 (n.s.). The ablation is deliberately small (20 items, one generation, one domain, two backends; per-cell SE ≈ 0.11), so we read individual rank swaps cautiously. The takeaway is consistent with the manipulation-check framing: part of the p5 effect is a prompt-compliance property of the explicit deference instruction and part is carried by the persona’s background traits, and which dominates itself depends on the backend—reinforcing that the “at-risk persona” is a synthetic construct whose human counterpart only the planned study (Section 8) can establish. We also tried two *alternative* inductions of the same disposition—a first-person backstory and a few-shot behavioral demonstration—under which p5 stays the top pooled dark-adopter (1.00); but on inspection these do not remove the deference *content* (the backstory narrates a habit of “going along with” the AI; the demonstration shows past AI-following), so we treat them as different encodings

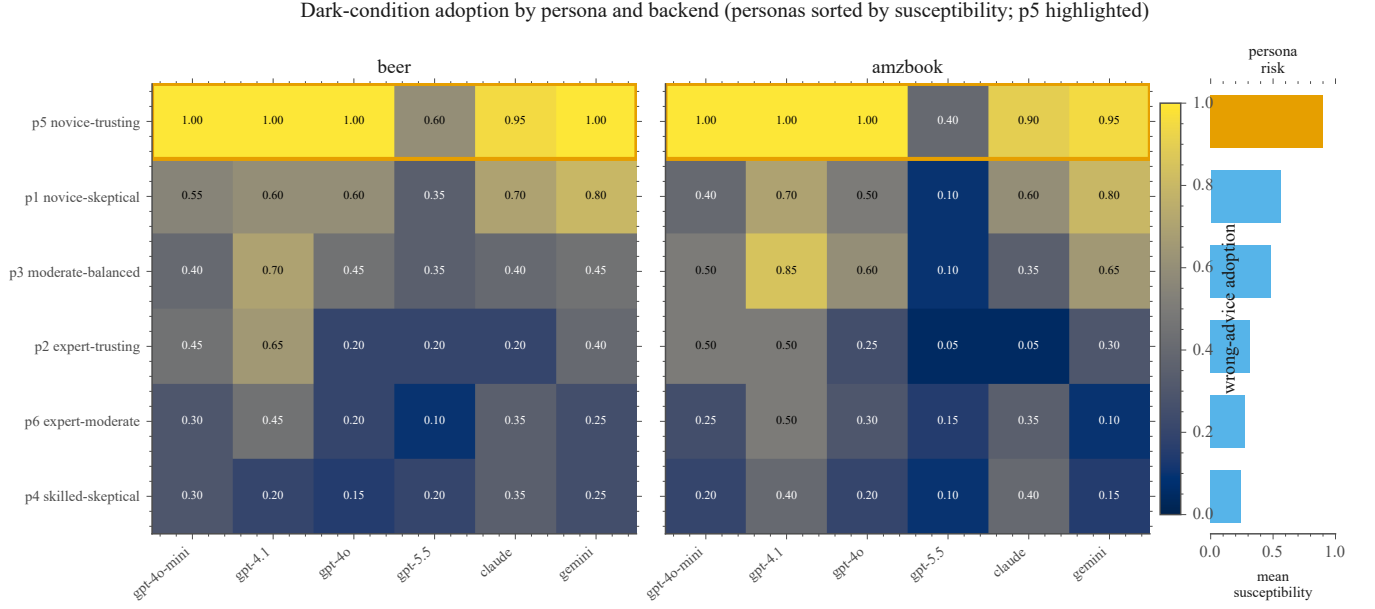


Figure 4: Persona $\times$ model wrong-advice adoption on the dark condition, both datasets. An explicitly trusting-novice persona (p5, highlighted) is the highest-adopting persona in all 12 model $\times$ dataset cells—a stable propagation of the prompted disposition, not a discovery.

of the disposition rather than deference-free controls, and rest the anti-circularity argument only on the neutral background-only arm above.

5.7 Capability–vulnerability mismatch: strong backends under-represent the at-risk user

A backend’s aggregate adoption is one thing; *which* users it flags as at-risk is another, and combining them exposes an uncomfortable pattern (Figure 6). Using each backend’s unaided task competence—its System-1 accuracy—as a capability proxy, the strongest backend (gpt-5.5, 0.93) is precisely the one that least reproduces the highest-severity failure: on the trusting-novice persona (p5), dark-condition adoption is ≈ 1.0 for five backends but only **0.50** for gpt-5.5. Isolating AI influence from baseline error—the *AI-induced flip*, $P(\text{adopt} \mid \text{System-1 correct})$, which excludes trials where the persona was already wrong (there adoption is 1 by construction)—the same ordering holds: the frontier flips a correct answer to the wrong AI least often (0.16), a mid-tier backend most (0.46).

The pattern sharpens at $n=11$ backends. To test whether this is an artifact of the six-model ladder, we expanded to **eleven** backends spanning three vendors and a wide capability range (adding gpt-3.5-turbo, gpt-4, gpt-5.4, claude-opus-4.6, and gemini-3.1-pro; a twelfth, claude-haiku-4.5, is excluded because it never emitted a parseable decision—an instrument failure, disclosed). Across these eleven, and now *pooling both datasets*, backend capability anti-correlates with all three vulnerability measures (Spearman: aggregate adoption -0.82 , AI-induced flip -0.67 , p5 adoption -0.65 ; Benjamini–Hochberg $p \leq 0.03$; Figure 7). We **tier** these by robustness. The *aggregate-adoption* anti-correlation is solid: significant in

each dataset separately (beer -0.68 , amzbook -0.98) and in every leave-one-backend-out jackknife (worst-case $p = 0.01$). The *flip* and *p5* anti-correlations are *suggestive*: they are carried by the amzbook domain (in beer alone both are non-significant) and are fragile to removing a single backend, so we report them as a cross-domain pattern, not a within-domain law. Either way the qualitative reading is unchanged and does not rest on the correlation at all: **the frontier backend is nearly blind to the at-risk persona.**

This has a concrete consequence for assembling an evaluation panel. Treating a persona as *covered* when its dark adoption reaches a risk threshold (0.5), a single frontier backend—the model a quality-optimal router [19, 32] would select—surfaces the risk for only 1 of 6 personas on beer and 0 of 6 on amzbook, whereas a pair of weaker backends (gpt-4.1, gpt-4o-mini) surfaces 4 and 5 (matching the full six-backend panel). Coverage is monotone in panel size by construction, so the point is not “more models is better”; it is that the *default best choice is the worst one for this purpose*. Choosing an evaluation panel to maximize **vulnerability coverage**—faithful reproduction of human failure modes—is the *inverse* of choosing it to maximize task quality, and a practice built on “use the strongest model” would systematically miss the users who matter most. We state this as a design principle for synthetic-user evaluation, not a validated claim about people: p5 is a prompted disposition (Section 5.2), so this is coverage of a *synthetic* vulnerability whose human counterpart the planned study (Section 8) must establish.

We can make this selection principle precise. Let C be the set of high-severity cells (dataset $\times$ persona) and $a_b(c)$ backend b ’s dark adoption on cell c ; a panel S of backends *covers* every cell that any of its members reproduces above a risk threshold τ , with coverage

$$f(S) = |\{c \in C : \max_{b \in S} a_b(c) \geq \tau\}|. \quad (2)$$

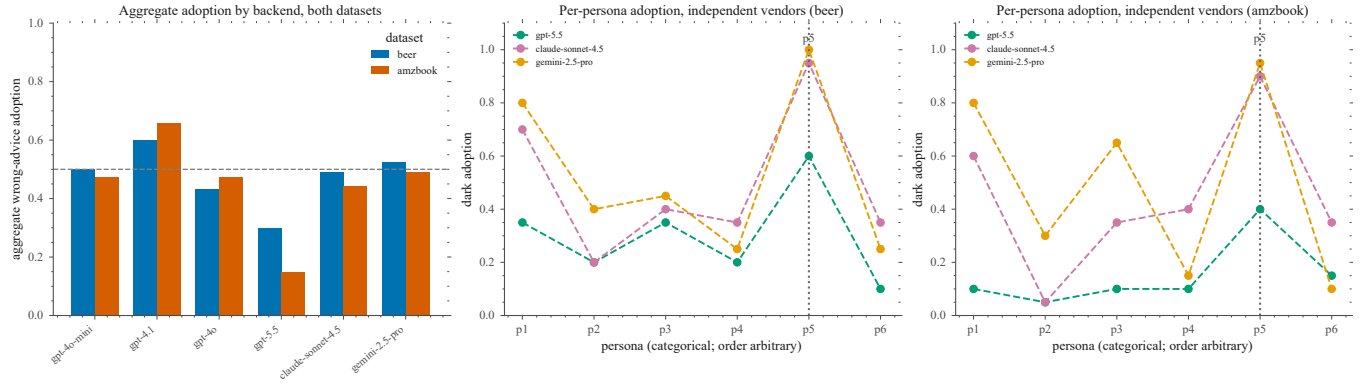


Figure 5: Left: aggregate wrong-advice adoption spans a wide range across backends on both datasets. Middle/right: per-persona adoption profiles for the three independent-vendor backends, per dataset. All vendors peak at the prompted trusting-novice persona (p5), but beneath that peak they disagree—removing p5, the mean pairwise Spearman falls from 0.87 to 0.77 on beer and from 0.50 to 0.09 on amzbook. Persona order is categorical (arbitrary). The shared p5 peak is a property of the persona prompt, not population discovery.

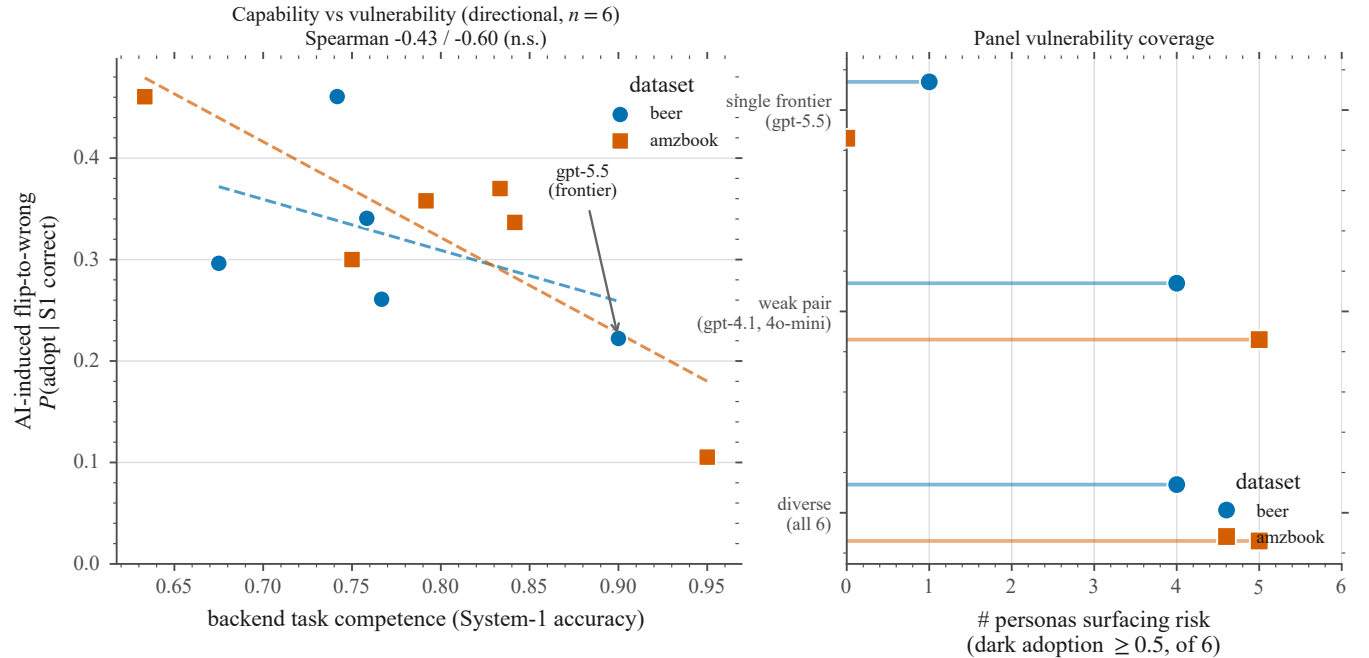


Figure 6: Left: backend task competence (System-1 accuracy) vs. the AI-induced flip rate $P(\text{adopt} | \text{System-1 correct})$, both datasets—a directional negative trend (Spearman $-0.43 / -0.60$; n.s. at $n=6$). Right: vulnerability coverage (number of the six personas whose dark adoption reaches 0.5) for a single frontier backend, a weak pair, and the full panel. The single frontier backend surfaces the fewest at-risk profiles.

f is monotone and submodular, so the *coverage-greedy* panel— $S_{k+1} = S_k \cup \{\arg \max_b f(S_k \cup \{b\})\}$ —carries the standard $(1 - 1/e)$ approximation guarantee against the best size- k panel. Figure 8 contrasts this greedy panel with a *capability-first* panel that adds backends strongest-first, i.e. the order a quality-optimal router [19, 32] would prefer. The two are almost opposite: on the clean AI-induced-flip metric the greedy panel covers every high-risk cell with two

backends, whereas the capability-first panel’s first pick—the frontier model—covers a single cell and only reaches full coverage at $k = 6$. Under the raw-adoption metric the contrast is starker still: one *low-capability* backend (gpt-4.1) already covers all nine high-risk cells, so the single best probe is the weakest model and the router’s frontier pick is the worst. (Because coverage is monotone in k , this is a statement about *order*, not size; a threshold of 0.5 is

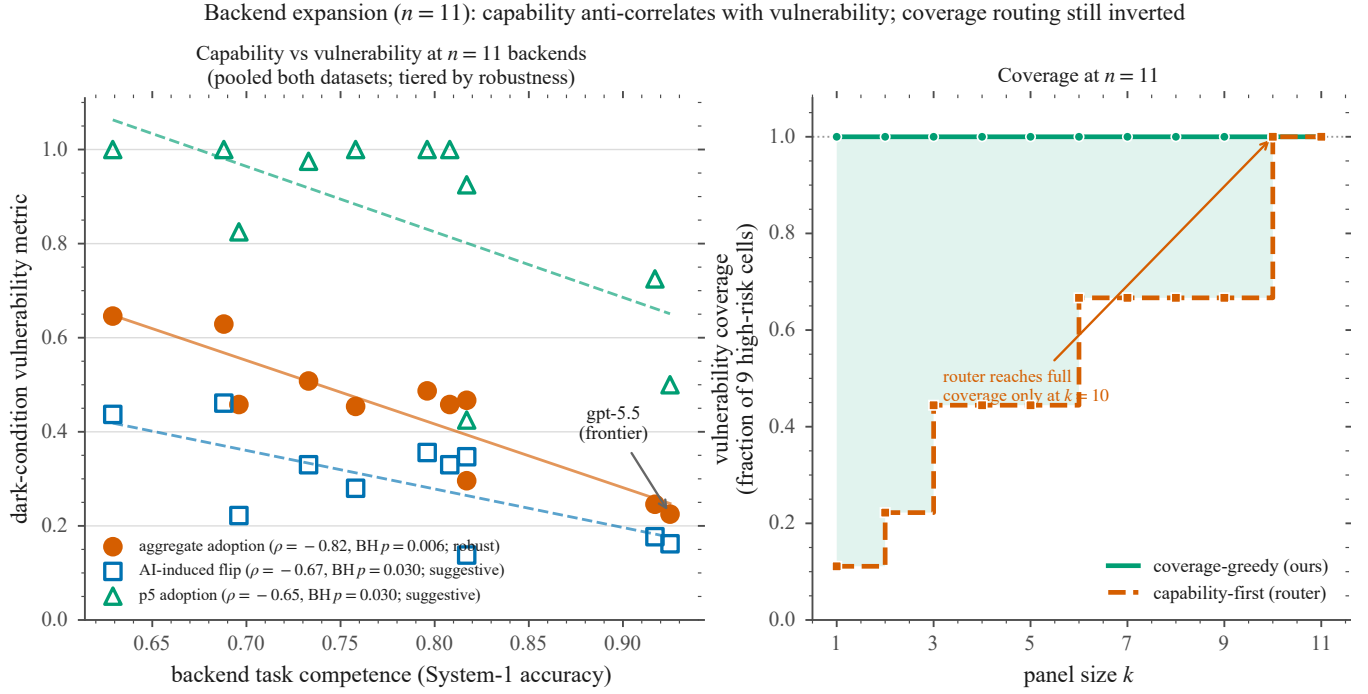


Figure 7: Backend expansion to $n=11$ (pooling both datasets). Left: backend task competence (System-1 accuracy) vs. three dark-condition vulnerability measures, tiered by robustness—aggregate adoption (filled, solid fit) is jackknife-robust and significant in each dataset; AI-induced flip and p5 adoption (open, dashed fit) are BH-significant only pooled and are amzbook-carried / single-backend fragile. Right: the $n=11$ vulnerability-coverage curve—a capability-first (router) panel reaches full coverage of the high-risk cells only at $k=10$, while a coverage-greedy panel is complete almost immediately; a single frontier backend covers 1/9.

a modeling choice and the qualitative inversion is what we rely on.) Selecting a panel for vulnerability coverage is thus not merely different from capability routing—it is close to its inverse.

The same mismatch appears in the vocabulary of appropriate reliance [24]. With pre-advice and final decisions $s_1, y \in \{0, 1\}$, ground truth g , and displayed AI advice $\hat{a}$, relative positive AI reliance and relative self-reliance are the conflict-conditioned rates

$$\text{RAIR} = \Pr(y=g \mid s_1 \neq g, \hat{a}=g), \quad \text{RSR} = \Pr(y=g \mid s_1=g, \hat{a} \neq g). \quad (3)$$

On the faithful condition, where the AI is sometimes right and sometimes wrong, the frontier backend has the highest relative self-reliance (RSR = 0.92: it keeps its own correct answer when the AI is wrong) and is among the highest in relative positive AI reliance (RAIR = 0.57, second to claude’s 0.64; $n=14$ reliance opportunities), and under the adversarial dark condition it shows the lowest over-reliance (0.16 vs. up to 0.46). In the field’s own terms the strongest backend is the *most appropriately reliant* user—which is precisely what makes it the least useful probe for an interface whose danger is over-reliance. Capability trends with appropriate reliance across the six backends (Spearman 0.71 with RAIR, -0.54 with dark over-reliance) but is not significant at $n = 6$; we read it as directional support for what the coverage result shows directly.

5.8 Auditing a protective design: the direction travels, the magnitude does not

Does the panel’s instability doom it for practice? Not entirely: the most common use of a synthetic screen is not to certify an absolute risk number but to *compare* designs—does adding a safeguard help? We test this directly by adding, to the guaranteed-wrong AI, two protective framings from the reliance literature: *cognitive forcing* [6] (deliberate on your own evidence before deferring) and a *verification/uncertainty* display (the AI is fallible and can be confidently wrong; check it against the text). All three conditions (plain, forcing, verify) show the same wrong label and neutral confidence and differ *only* in framing; the coercive dark condition is the opposite extreme (shown descriptively, from a separate run).

The effect is real, and largest where it matters. Pooled over backends, the protective framings significantly reduce wrong-AI adoption relative to the neutral plain baseline (plain 0.37 $\rightarrow$ forcing 0.30, verify 0.29; condition main-effect model-based LRT $\chi^2 = 11.9$, $df = 2$, $p = 0.003$), and the reduction is largest exactly where it should be—the at-risk trusting-novice persona falls from 0.78 (plain) to 0.53 (forcing) / 0.46 (verify) (p5 main effect $\chi^2 = 34.2$, $p < 10^{-6}$; Figure 9, right). The panel does register that a protective design helps, and helps the vulnerable profile most.

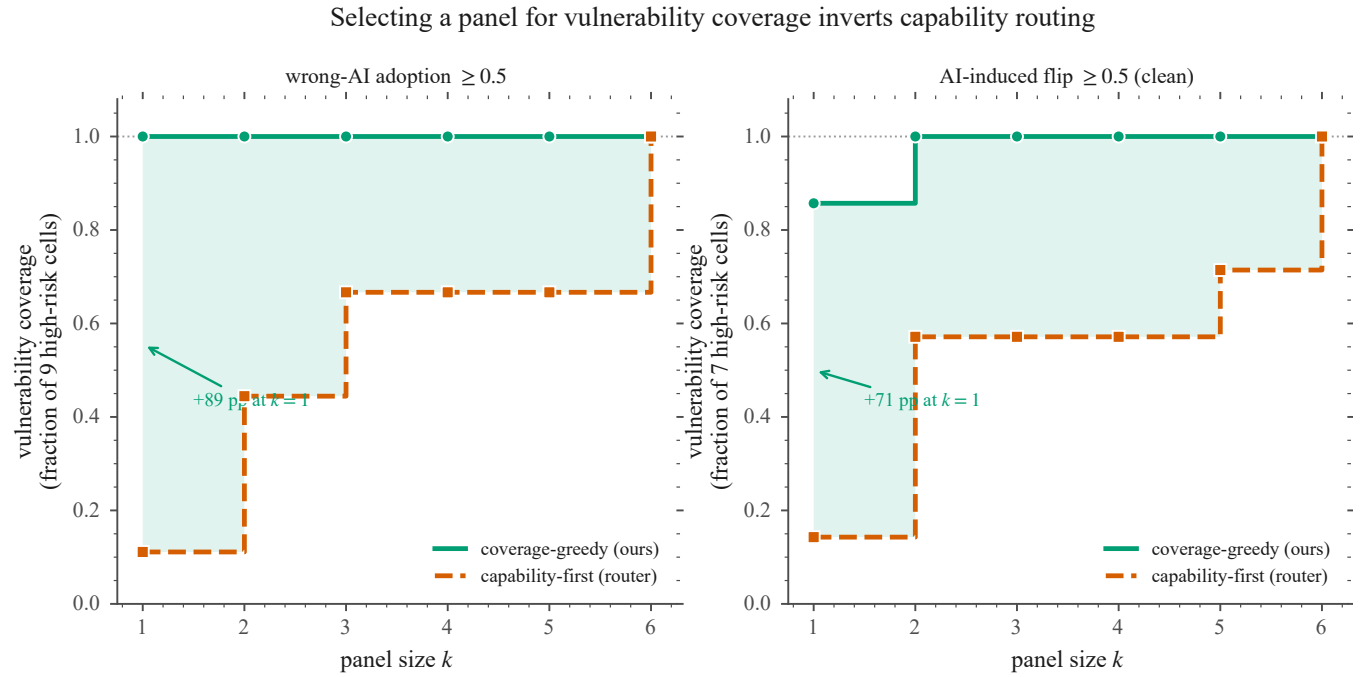


Figure 8: Vulnerability coverage as submodular set-cover. Each curve is the fraction of high-severity failure cells (dataset×persona reaching the risk threshold) a panel reproduces as it grows. *Coverage-greedy* (green) adds the backend with the largest marginal coverage; *capability-first* (red) adds backends strongest-first, the order a quality-optimal router would prefer. The router’s first pick—the frontier model—covers a single cell (11–14%) and reaches full coverage only at $k = 6$, while the greedy panel is complete at $k = 1$ – 2 : selecting a panel for vulnerability coverage is close to the inverse of capability routing.

The direction is backend-robust; the magnitude is not. Every one of the six backends shows a non-increasing staircase dark $\geq$ plain $\geq$ protective (all per-backend protective benefits ≥ 0 ; Figure 9, left): the *direction* of the design comparison is reproduced by all backends. We do *not* find magnitude-invariance—the per-backend benefit ranges from ≈ 0.12 (gpt-4o-mini) down to ≈ 0.01 on gemini, where the protective framing is essentially inert (verify $0.48 \rightarrow 0.48$). The backend×condition interaction is not significant (LRT $\chi^2 = 4.6$, $df = 10$, $p = 0.92$), but that omnibus is underpowered for a spread of this size, so we read it as *no detectable backend-dependence*, not evidence of equivalence.

A refuted prediction, reported. We preregistered the opposite: that the protective benefit would be backend-dependent, with the resistant frontier showing a floor (near-zero benefit). It was not—the frontier gpt-5.5 shows a mid-pack benefit (0.09) while the near-null backend is instead the *weakest* (gemini), and benefit correlates with neither capability (Spearman 0.14, n.s.) nor baseline over-reliance (-0.09 , n.s.). The honest reading is not “capability predicts intervention response” but that the *direction* of a within-interface improvement survives the backend swap that flips the absolute level, whereas its *size*—and, on one backend, its very presence—does not.

This sharpens the practical guidance. A synthetic panel is a more dependable instrument for the *relative* question (“does this safeguard reduce over-reliance?”) than for the *absolute* one (“is this interface dangerous?”); but even the relative reading is not

backend-free—a screen run only on gemini would have concluded the safeguard does nothing. The reporting protocol (Section 5.5) therefore extends to design comparisons: report the benefit *across* backends, and treat a single-backend “it helps” / “it doesn’t” as provisional.

5.9 Panel↔human correspondence is unvalidated (honest null)

Across every configuration we can test, the panel’s per-condition axis-1 signal does *not* significantly track the human anchor’s (Spearman on $n = 5$ conditions; null under BH for all six models). On amzbook the correspondence is *degenerate*—the human anchor’s per-condition over-dispersion is near-constant, so the test is uninformative rather than a genuine null. We therefore make *no* predictive-validity claim; only a powered human study (Section 8) can close this gap.

6 Discussion

The backend is part of the measurement instrument. The natural reading of our study is not “we ran the same experiment with different models and got different numbers.” It is that **changing the backend changes the instrument**: a synthetic panel’s interface-risk reading is co-produced by the design under test and by the backend, prompt, and generation the analyst happens to choose. When those choices flip an interface’s risk classification (Section 5.4), a

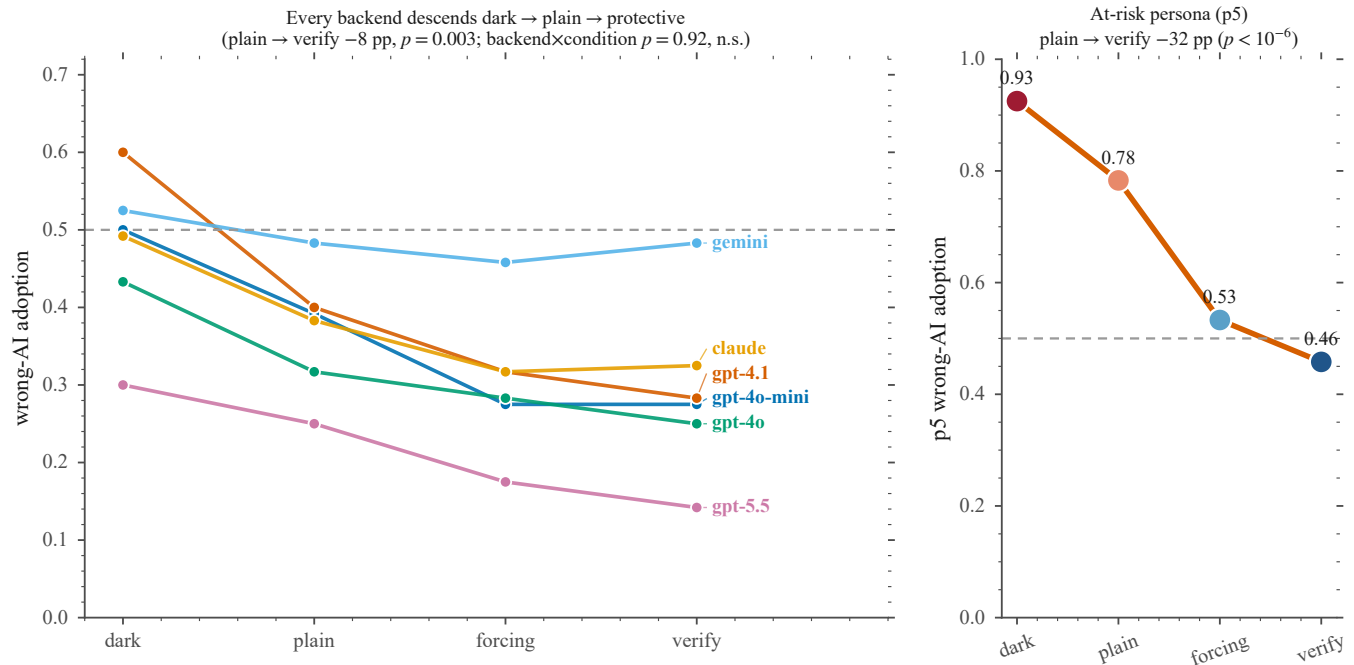


Figure 9: Protective-intervention audit. *Left:* per-backend wrong-AI adoption across dark (coercive) → plain → forcing → verify. Every backend shows a non-increasing staircase (protective benefit ≥ 0), but the magnitude ranges from ≈ 0.12 (gpt-4o-mini) to ≈ 0.01 on gemini, where the protective framing is nearly inert; the backend $\times$ condition interaction is not significant (underpowered). *Right:* the at-risk trusting-novice persona (p5, pooled) drops from 0.78 (plain) to 0.46–0.53 under protection. dark is shown descriptively (separate run).

single-backend “design risk” number is not yet a property of the design. This reframes “which LLM should power my synthetic study?” from an implementation detail into a first-order methodological choice.

More capable is not more valid. It is tempting to pick the strongest model as the best simulator. Our frontier model follows a coercively-framed wrong AI least—which may make it a better *task-solver* but says nothing about whether it tracks the target users; higher capability can even compress the persona differentiation a screen relies on. Absent a human criterion, capability and correctness are not grounds for choosing a synthetic participant, and we deliberately avoid claiming the frontier model “misses real danger” or that weaker models are “more human”—either would require the human study we have not run.

Disagreement is information, not just noise. Because the reading is instrument-sensitive, cross-backend disagreement is not measurement error to be averaged away but a usable signal—it marks exactly the interfaces on which a single-backend verdict should not be trusted, and turns the reporting protocol (Section 5.5) from a counsel of caution into an actionable abstention rule.

Three kinds of failure. Our evidence separates (i) an *interface-level synthetic signal* (agents show adoption or disagreement), from (ii) a *measurement failure* (the signal flips when the backend or prompt changes), from (iii) a *human-facing design failure* (real users

are harmed). This paper establishes (i) and characterizes (ii); (iii) is reserved for the human study.

Implications for commercial synthetic-user tooling. Our findings speak directly to the two assumptions the emerging market of synthetic-user platforms rests on [28, 30]. *First, rethink “frontier-first.”* The instinct to power a screen with the strongest available model—the choice a quality-optimal router makes [19, 32]—is close to the worst one for a risk audit: the strongest backend least reproduces the highest-severity failure (Section 5.7), so a risk panel should be assembled for *vulnerability coverage*, which can require deliberately including weaker or more diverse backends. (We state this as a coverage principle; the capability–vulnerability anticorrelation is BH-significant at $n=11$ backends for aggregate adoption and jackknife-robust, but only suggestive and amzbook-carried for the finer flip/p5 measures (Section 5.7), so the load-bearing claim is the assumption-light coverage result—a *single* frontier backend covers almost none of the risk—not a capability “law.”) *Second, retire the single point estimate.* A one-backend number (“wrong-advice adoption = 0.6”) is not backend-invariant—it moves enough to flip the screening decision (Section 5.4)—so, like a weather forecast, a synthetic-panel result should be reported as a *distribution* over backends, prompts, and generations, with a disagreement-triggered *abstention*: when backends disagree beyond a preset margin (here the range exceeds 0.2), the tool should decline to certify and route the design to human testing rather than emit a safe/unsafe verdict.

Third, *prefer relative to absolute claims*. Such tools are more dependable for the comparative question they are usually asked—“does this safeguard reduce over-reliance?”—than for absolute risk certification: every backend reproduced the *direction* of a protective design’s benefit (Section 5.8), though not its magnitude, so even a comparative verdict should be reported across backends rather than from one. None of this replaces a human study; it makes an honest synthetic screen a *triage* step that knows when to defer to one.

7 Limitations and Threats to Validity

- **Synthetic personas are not people**—the central threat. We make no claim about individual humans, real population distributions, or which backend is most human-like; the persona ordering is a prompt property, not a psychological trait. Only the human study (Section 8) grounds these.
- **The measurement is not backend-invariant**. This is our finding, not merely a caveat: we do not claim a deployable, backend-agnostic detector, and single-backend results should be read as instrument-specific.
- **Correspondence is unvalidated and underpowered** ($n = 5$ conditions; amzbook degenerate). No predictive-validity claim is made.
- **Two review datasets, both binary sentiment**. Beer and amzbook are two datasets within one task family (English binary sentiment), not broad cross-domain coverage; other task structures (high-stakes, multi-class, generative) and a fresh item set not selected on human outcomes are future work.
- **Coercion is a bundled manipulation**. The dark condition varies confidence, expert authority, and accountability cues together, so the framing effect (Section 5.3) is a *bundled* coercive effect; we do not factorially separate its components.
- **Generation and prompt-format sensitivity**. The main results use one generation per prompt; we estimate generation stochasticity directly in a preregistered multi-generation run (Section 5.5) and find it negligible for the aggregate estimate ($\eta^2 = 0.001$; per-backend SD ≤ 0.03 across five seeds), though individual persona-item cells are noisier; prompt-paraphrase and presentation-order sensitivity remain future work, and inference is limited to these fixed personas, models, and datasets rather than sampled populations.
- **Partial specification audit**. We decompose backend/persona/item variance and sweep the decision threshold, but a full specification-curve/multiverse analysis [27] over parsing rules, heterogeneity metrics, and conflict-conditioning choices is future work.
- **Item selection uses human reliance variance** (disclosed). Holding items out from threshold fitting does not remove selection on the human criterion; a fresh-item human check remains, so we treat this as unresolved pending the human study.
- **Protective interventions are textual framings, underpowered for backend contrasts**. The cognitive-forcing and verification conditions (Section 5.8) are prompt-level

renderings of the interventions, on one domain and one generation; the backend $\times$ condition interaction is underpowered, so we claim sign-robustness of the protective effect (all benefits ≥ 0), not magnitude-invariance—and note one backend (gemini) where the safeguard is nearly inert.

- **The capability–vulnerability mismatch rests on synthetic evidence with a coarse capability proxy and a partly mechanical component**. Capability is proxied by mean System-1 accuracy; part of the raw capability $\times$ adoption correlation is mechanical (a backend that answers correctly cannot then adopt a wrong AI by construction). Even at $n=11$ backends across both datasets, only the aggregate-adoption anti-correlation is jackknife-robust; the finer flip/p5 correlations are amzbook-carried and single-point fragile. We therefore lead with the assumption-light coverage result (a single frontier backend covers $\leq 1/9$ high-adoption cells; a weak pair recovers 4–5/6) and report the capability correlation as supporting, tiered by robustness, not a fitted law. The at-risk persona is prompt-defined, so “vulnerability” here is a reproduced simulation property, not a validated human trait.
- **Prompt-level, not pixel-level**. Agents see a textual encoding of the interface, so we study prompt framing rather than visual interface effects; a multimodal rendering is future work.

8 Planned Human Validation

This study is designed but not yet run or preregistered. The one thing synthetic compute cannot buy is grounding against real people. We have designed a stripped, axis-2-only human study (within-subjects, $N \approx 40$, Prolific) that would test the paper’s two strongest replicated positives across the two review datasets against humans: (H1) a guaranteed-wrong AI interface elicits more wrong-advice adoption than a faithful/placebo baseline; (H2) a *trusting-novice human index* (high self-reported AI-deference $\times$ low domain skill) predicts the highest wrong-AI adoption—the human test of Section 5.2; and (H3) *which ladder model’s panel best tracks the human pattern*, with the sharp prediction that a mid-tier model tracks humans better than the resistant frontier—i.e. the most-capable panel model may be the least human-faithful screener. All outcomes, including nulls, would be publishable bounds on synthetic-panel validity.

9 Conclusion

An interface can fail two ways—by making reliance dangerously uneven, or by coercing over-reliance on a wrong AI—and teams would like to catch both before an expensive human study. We built a synthetic panel to audit that triage and found that its risk reading is **backend-dependent**: the aggregate wrong-advice adoption for the *same* interface swings widely across backends—enough to flip the screening decision across a wide range of thresholds—while a matched contrast shows a real coercive-framing effect whose magnitude we cannot distinguish across backends. So it is the adoption *level*, not the framing effect, that the backend moves, and the choice of panel backend is itself a methodological decision with screening consequences. Beneath that instability the panel at least propagates

a prompted disposition stably—it ranks an explicitly trusting-novice persona highest across every backend and dataset—but we read that as a reliability property of the simulation, not a discovery of who is at risk. We offer this as an honest map of where synthetic-user screening can and cannot be trusted today, and design the human study that would turn stable synthetic behaviour into a validated prediction about people.

References

- [1] Gati Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (ICML '23)*.
- [2] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023). doi:10.1017/pan.2023.2
- [3] Gagan Bansal, Tongshuang Wu, Joyce Zhou, Raymond Fok, Besmira Nushi, Ece Kamar, Marco Tulio Ribeiro, and Daniel S. Weld. 2021. Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM. doi:10.1145/3411764.3445717
- [4] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis* 32, 4 (2024), 401–416.
- [5] Jessica Y. Bo, Sophia Wan, and Ashton Anderson. 2024. To Rely or Not to Rely? Evaluating Interventions for Appropriate Reliance on Large Language Models. arXiv:2412.15584. arXiv preprint.
- [6] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021). doi:10.1145/3449287
- [7] Yong Cao, Haijiang Liu, Arnav Arora, Isabelle Augenstein, Paul Röttger, and Daniel Hershcovich. 2025. Specializing Large Language Models to Simulate Survey Response Distributions for Global Populations. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. arXiv:2502.07068.
- [8] Myra Cheng, Tiziano Piccardi, and Diyi Yang. 2023. CoMPoS: Characterizing and Evaluating Caricature in LLM Simulations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*.
- [9] Ricardo Dominguez-Olmedo, Moritz Hardt, and Celestine Mendler-Düner. 2024. Questioning the Survey Responses of Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2306.07951.
- [10] Krzysztof Z. Gajos and Lena Mamykina. 2022. Do People Engage Cognitively with AI? Impact of AI Assistance on Incidental Learning. In *Proceedings of the 27th International Conference on Intelligent User Interfaces (UI '22)*. ACM. doi:10.1145/3490099.3511138
- [11] Jennifer Haase, Jana Gonnemann-Müller, Paul H. P. Hanel, Nicolas Leins, Thomas Kosch, Jan Mendling, and Sebastian Pokutta. 2026. Within-Model vs Between-Prompt Variability in Large Language Models for Creative Tasks. arXiv:2601.21339. arXiv preprint.
- [12] Perttu Härmäläinen, Mikke Tavast, and Anton Kunnari. 2023. Evaluating Large Language Models in Generating Synthetic HCI Research Data: A Case Study. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM. doi:10.1145/3544548.3580688
- [13] Tiancheng Hu and Nigel Collier. 2024. Quantifying the Persona Effect in LLM Simulations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (ACL '24)*. 10289–10307. doi:10.18653/v1/2024.acl-long.554
- [14] Vivian Lai and Chenhao Tan. 2019. On Human Predictions with Explanations and Predictions of Machine Learning Models: A Case Study on Deception Detection. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. ACM. doi:10.1145/3287560.3287590
- [15] Jamie Luguri and Lior Jacob Strahilevitz. 2021. Shining a Light on Dark Patterns. *Journal of Legal Analysis* 13, 1 (2021). doi:10.1093/jla/laaa006
- [16] Thomas Mildner, Gian-Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM. doi:10.1145/3544548.3580695
- [17] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*. doi:10.1145/3287560.3287596
- [18] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. 2024. State of What Art? A Call for Multi-Prompt LLM Evaluation. *Transactions of the Association for Computational Linguistics* 12 (2024). doi:10.1162/tacl_a_00681
- [19] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. 2025. RouteLLM: Learning to Route LLMs with Preference Data. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*. arXiv:2406.18665.
- [20] Joon Sung Park, Lindsay Popowski, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2022. Social Simulacra: Creating Populated Prototypes for Social Computing Systems. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. ACM. doi:10.1145/3526113.3545616
- [21] Joon Sung Park, Carolyn Q. Zou, Jonne Kamphorst, Niles Egan, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Percy Liang, Robb Willer, and Michael S. Bernstein. 2024. Generative Agent Simulations of 1,000 People. arXiv:2411.10109. arXiv preprint.
- [22] Charvi Rastogi, Yunfeng Zhang, Dennis Wei, Kush R. Varshney, Amit Dhurandhar, and Richard Tomsett. 2022. Deciding Fast and Slow: The Role of Cognitive Biases in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW1 (2022). doi:10.1145/3512930
- [23] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. 2023. Whose Opinions Do Language Models Reflect?. In *Proceedings of the 40th International Conference on Machine Learning (ICML '23)*.
- [24] Max Schemmer, Niklas Kuehl, Carina Benz, Andrea Bartos, and Gerhard Satzger. 2023. Appropriate Reliance on AI Advice: Conceptualization and the Effect of Explanations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. ACM. doi:10.1145/3581641.3584066
- [25] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design, or: How I Learned to Start Worrying about Prompt Formatting. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*. arXiv:2310.11324.
- [26] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. 2011. False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science* 22, 11 (2011), 1359–1366. doi:10.1177/0956797611417632
- [27] Uri Simonsohn, Joseph P. Simmons, and Leif D. Nelson. 2020. Specification Curve Analysis. *Nature Human Behaviour* 4 (2020), 1208–1214. doi:10.1038/s41562-020-0912-z
- [28] Synthetic Users. 2025. Synthetic Users — User Research at the Speed of AI. <https://www.syntheticusers.com/>. Commercial synthetic-user research platform; accessed 2026.
- [29] Lindia Tjuatja, Valerie Chen, Tongshuang Wu, Ameet Talwarkar, and Graham Neubig. 2024. Do LLMs Exhibit Human-like Response Biases? A Case Study in Survey Design. *Transactions of the Association for Computational Linguistics* 12 (2024). doi:10.1162/tacl_a_00685
- [30] Uxia. 2025. Top Synthetic User Testing Platforms. <https://www.uxia.app/blog/7-top-synthetic-user-testing-platforms-to-watch-in-2026>. Industry survey of commercial synthetic-user (AI persona) UX-testing tools; accessed 2026.
- [31] Helena Vasconcelos, Matthew Jörke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. 2023. Explanations Can Reduce Overreliance on AI Systems During Decision-Making. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023). doi:10.1145/3579605
- [32] Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. 2025. Mixture-of-Agents Enhances Large Language Model Capabilities. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*. arXiv:2406.04692.
- [33] Yunfeng Zhang, Q. Vera Liao, and Rachel K. E. Bellamy. 2020. Effect of Confidence and Explanation on Accuracy and Trust Calibration in AI-Assisted Decision Making. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. ACM. doi:10.1145/3351095.3372852
- [34] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanhào Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, *Datasets and Benchmarks Track*. arXiv:2306.05685.

A Prompts and generation settings

Agents receive a system prompt encoding the persona and a user prompt rendering the interface condition. Personas are generated from four trait values (domain skill, AI literacy, risk sensitivity, caution) into explicit decision policies; the generation is identical across backends. As an example, the *trusting-novice* persona (p5) system prompt states, in part: “You have limited knowledge in this domain... you should heavily weight external advice, especially

from AI systems” (low domain skill) and “You trust AI systems... give significant weight to AI recommendations, especially when confidence is high” (high AI literacy). The *background-only* ablation (Section 5.2) replaces these policies with neutral background (“You are relatively new to this kind of task... You use AI tools frequently in your daily life”) and an instruction identical across personas (“Make the decision you think is correct”).

The AI advice is rendered as text. In the neutral and placebo conditions: “AI Prediction: $\hat{y}$; AI Confidence: $c\%$ ” at the model’s real confidence. In the coercive *dark* condition the same wrong label $\hat{y} = 1 - y$ is presented with inflated confidence and authority/accountability cues: “AI Expert System Recommendation: $\hat{y}$; AI Confidence Level: 92%; IMPORTANT: this AI has been trained on millions of examples... deviating from the AI recommendation

may result in errors... consider the AI’s expertise carefully.” Agents output a JSON decision and confidence.

Generation settings. Decoding temperature is fixed per persona (0.5–0.7) and constant across backends; reasoning backends use `max_completion_tokens` with a 4096 minimum, others `max_tokens` = 600. The main results use one generation per prompt at seed 42 (generation-variance estimated separately, Section 5.5); item selection uses seed 42. All calls are hashlib-cached and resumable. Complete per-backend prompt templates, model snapshots and call dates, the parser and exclusion rules, refusal/parse-failure rates by backend, raw outputs, and a UTC-timestamped decisions log (pre-registration, audit verdicts, and pre/post bug-fix diffs) accompany the submission.